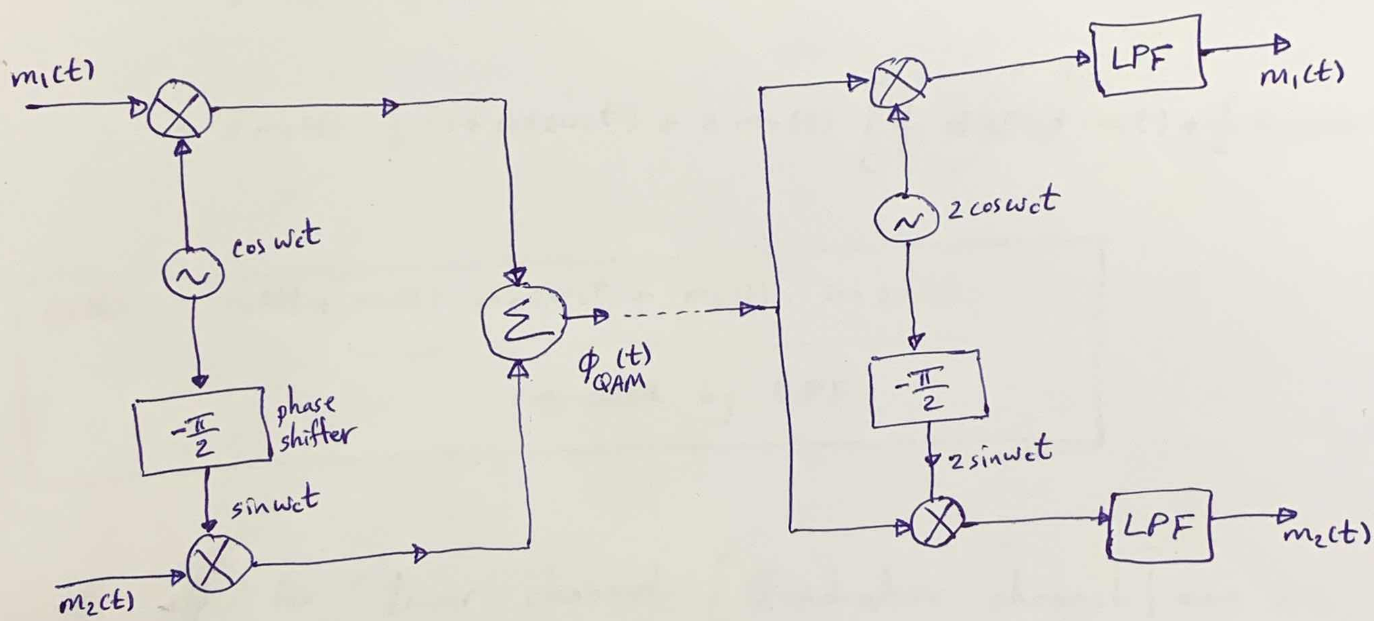
Quadrature Amplitude Modulation :- [QAM] "Quadrature Multiplexing".

- DSB signals occupy twice the B.W. required for the baseband.

⊛ Disadvantage ! What can we do ?

- transmitting two DSB signals using carriers of same frequency but in phase quadrature as shown below.



- the sum of two DSB-modulated signal is:-

$$\phi_{QAM}(t) = m_1(t)\cos w_c t + m_2(t) \cdot \sin w_c t .$$

- two baseband signals can be seperated at receiver by synchronous detection using two local carriers in phase quadrature.

- the o/p for upper channel [In-phase channel] can be written as :-
(I) channel.

$$x_1(t) = 2\cdot\phi_{QAM}(t) \cdot \cos w_ct = 2\left[m_1(t)\cdot\cos w_ct + m_2(t)\cdot\sin w_ct\right]\cdot\cos w_ct$$

$$= 2m_1(t)\cdot\cos^2 w_ct + 2\,m_2(t)\sin w_ct\cos w_ct$$

$$= \cancel{2}m_1(t)\cdot\frac{1}{\cancel{2}}(1+\cos 2w_ct) + 2\,m_2(t)\cdot\left[\frac{1}{2}\cancel{\sin(w_ct-w_ct)}^{0} + \frac{1}{2}\sin(w_ct+w_ct)\right]$$

$$\boxed{x_1(t) = m_1(t) + \underbrace{m_1(t)\cdot\cos 2w_ct + m_2(t)\cdot\sin 2w_ct}_{\text{suppressed by LPF.}}}$$

- the o/p for lower channel [Quadrature channel] can be written as :-
(Q) channel

$$x_2(t) = 2\cdot\phi_{QAM}(t)\cdot\sin w_ct = 2\left[m_1(t)\cdot\cos w_ct + m_2(t)\cdot\sin w_ct\right]\cdot\sin w_ct$$

$$x_2(t) = 2\, m_1(t) \cos w_c t \sin w_c t + 2 m_2(t) \cdot \sin^2 w_c t$$

$$= \cancel{2}\, m_1(t) \cdot \frac{1}{\cancel{2}} \left[\sin(w_c t + w_c t) - \sin(\cancel{w_c t - w_c t})^{0} \right] + \cancel{2} \cdot m_2(t) \cdot \frac{1}{\cancel{2}} (1 - \cos 2 w_c t).$$

$$x_2(t) = m_1(t) \cdot \sin 2 w_c t + m_2(t) - m_2(t) \cdot \cos 2 w_c t.$$

$$\boxed{x_2(t) = m_2(t) + \underbrace{m_1(t) \cdot \sin 2 w_c t - m_2(t) \cdot \cos 2 w_c t}_{\text{suppressed by LPF}}}$$

- Disadvantage :-

* slight error in phase or freq. of the carrier at the demodulator in QAM will leads to loss & distortion of signals & interference between two channels.

H.W Let the carrier at the demodulator be $2 \cdot \cos(w_c t + \underset{\substack{\uparrow \\ \text{phase error.}}}{\theta})$.

- check $x_1(t)$ & $x_2(t)$, what will happen? explain!!

- this cochannel interference is undesirable.

⊛ Cochannel refers to channels having same carrier frequency.

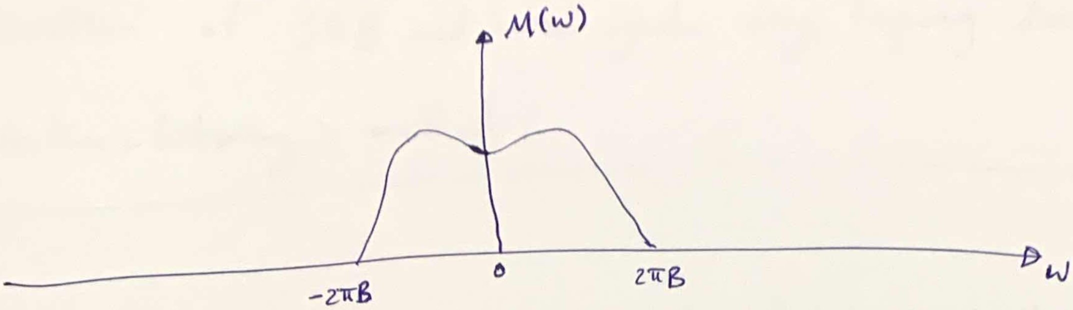
- In addition; unequal attenuation of USB & LSB during transmition also leads to crosstalk or cochannel interference.

AM Single SideBand (SSB)

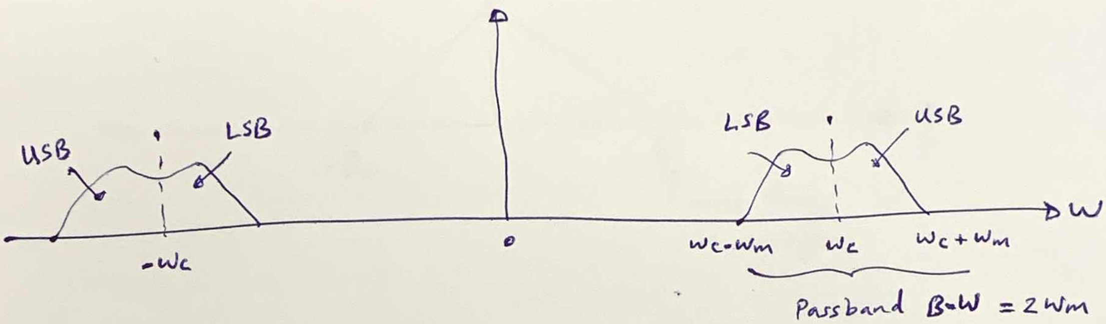
- the DSB spectrum has 2 sidebands, USB & LSB
- both of them contain complete information of the baseband signal.

- A scheme in which only one sideband is transmitted is known as SSB, which require $\frac{1}{2}$ the B.W of DSB signal.

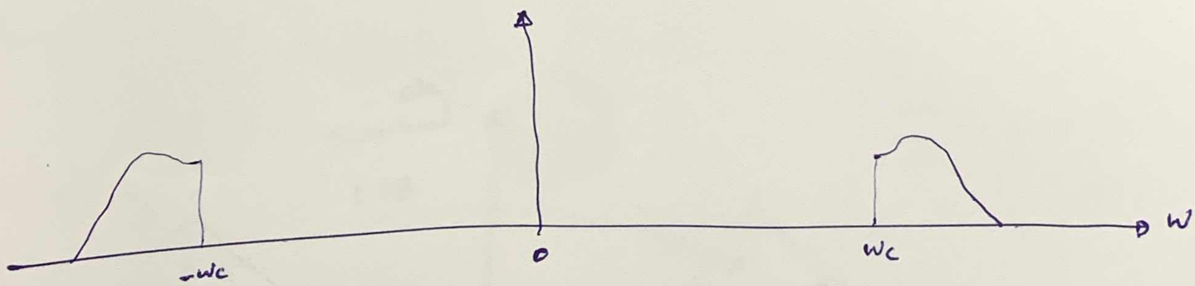
- An SSB signal can be coherently demodulated. As an example shown below.



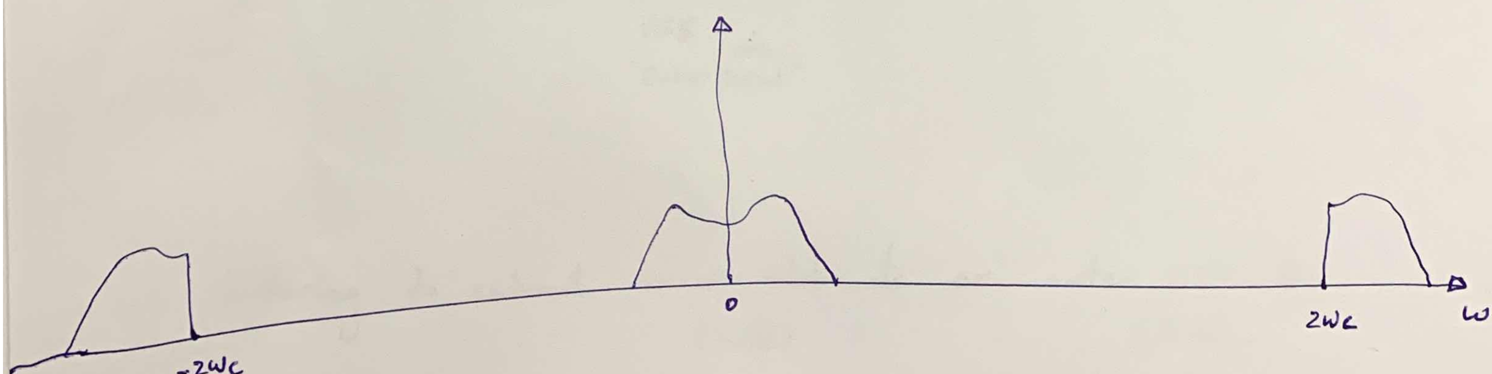
"Baseband"



"DSB"

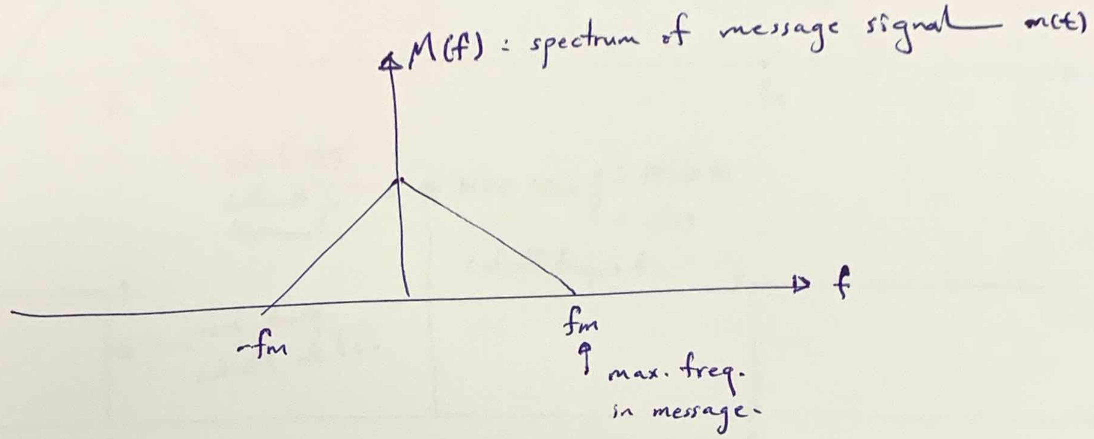


SSB : "USB"

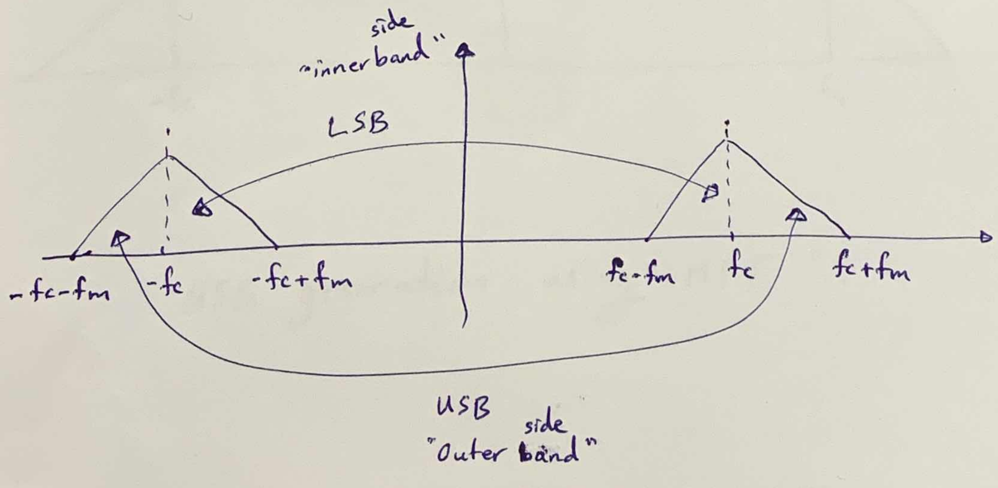


"SSB spectra".

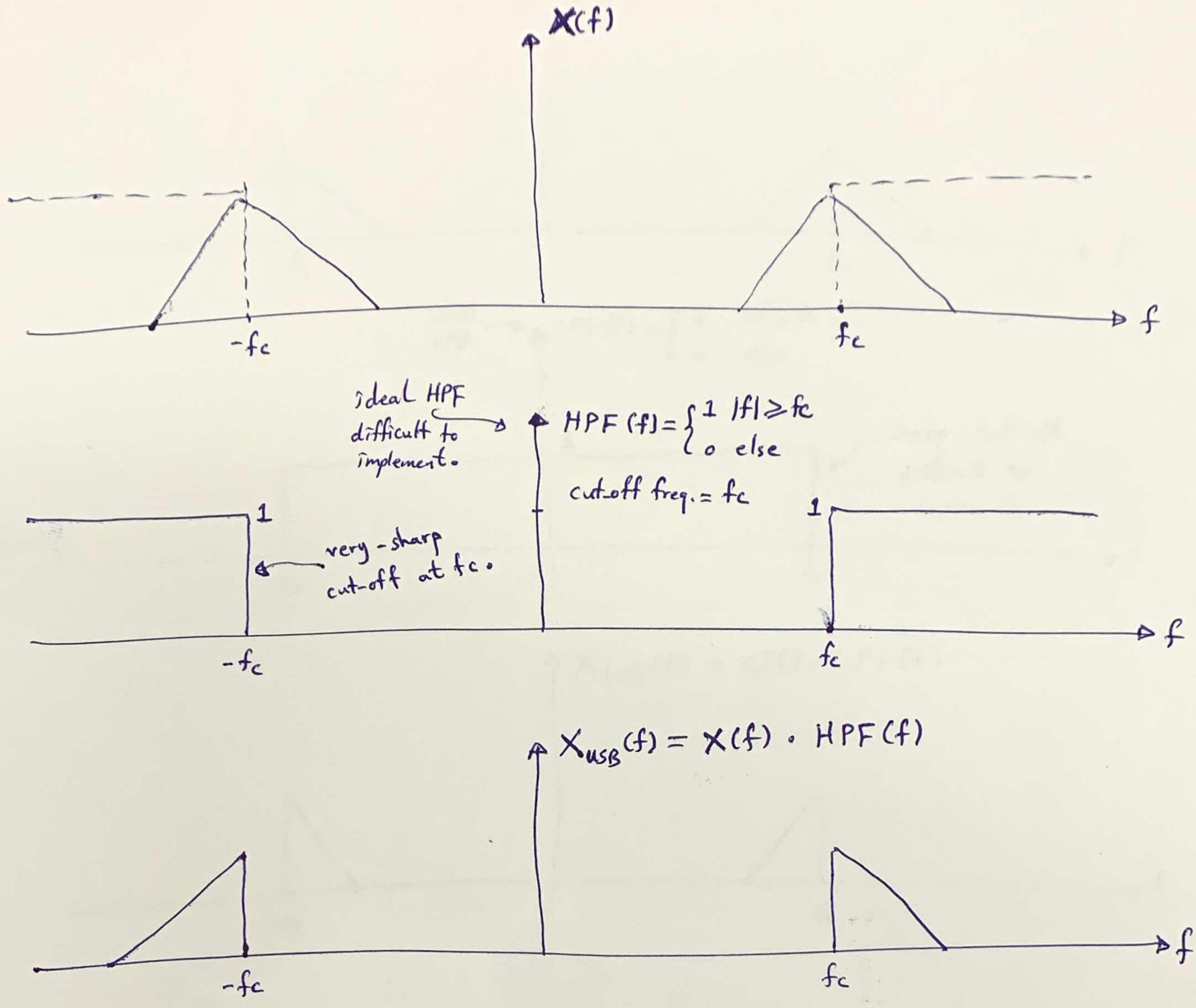
Generation of SSB modulated signal using frequency discrimination (selective - filtering) method :-



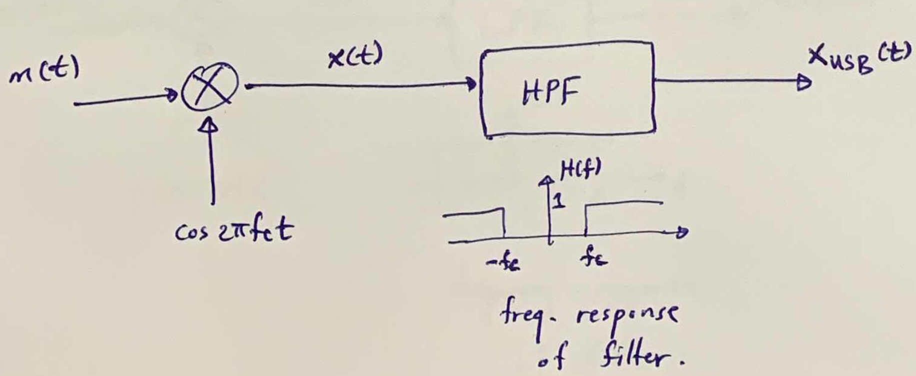
DSB - SC signal : $x(t) = m(t) \cdot \cos 2\pi f_c t$

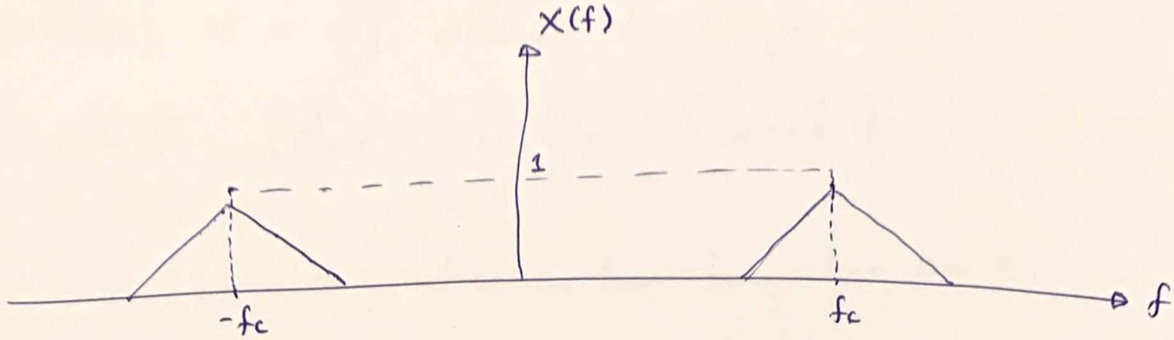


- use filtering to extract inner sidebands (LSB) or outer sidbands. (USB)

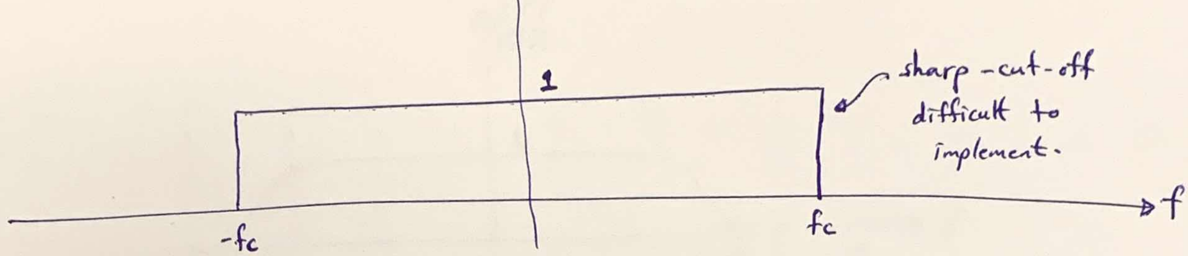


"USB generation using HPF".

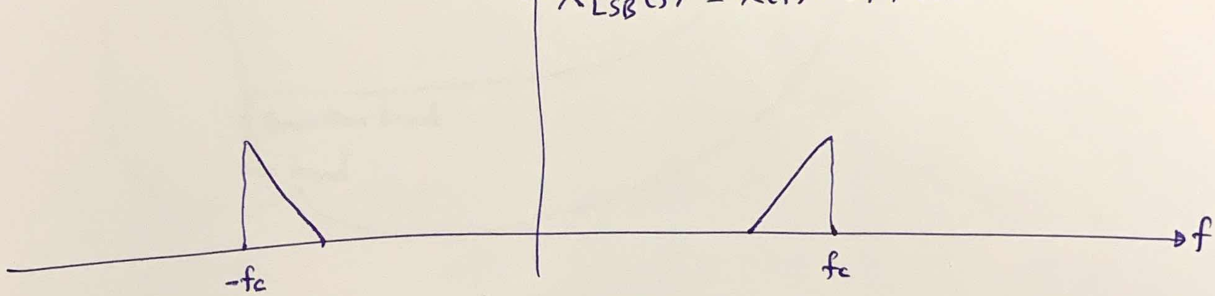




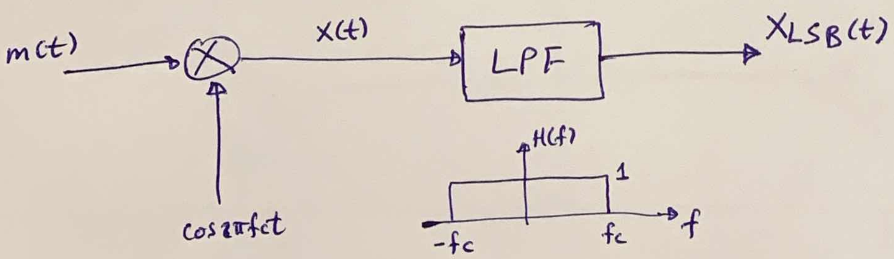
ideal LPF → $LPF(f) = \begin{cases} 1 & |f| \leq f_c \\ 0 & \text{else} \end{cases}$



$X_{LSB}(f) = X(f) \cdot LPF(f)$

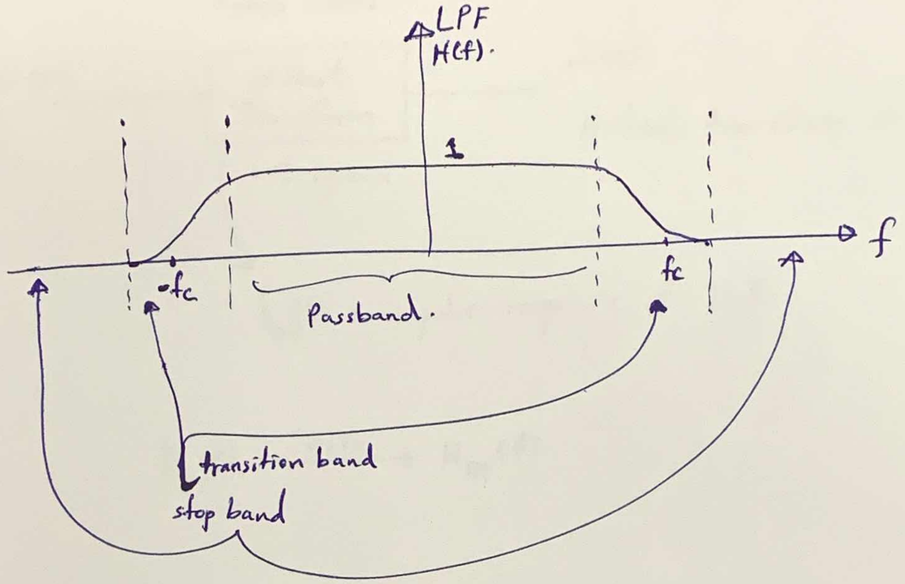


"LSB generation using LPF".

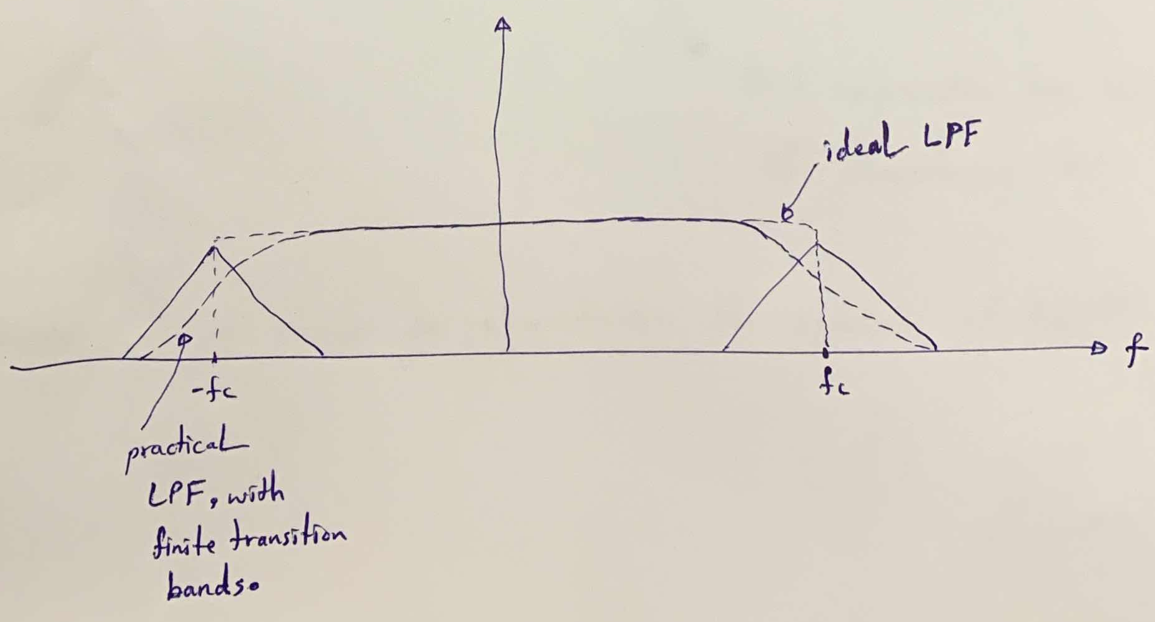


frequency-response of filter.

- in practice, it is very difficult to design a filter with a sharp-cut-off. (There is no transition band here, just passband & stop band, where in practice we have transition band).

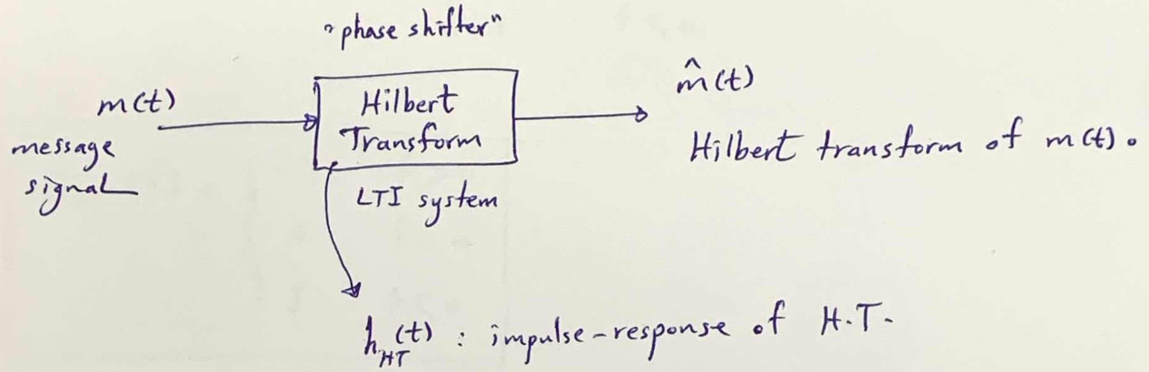


- practical filters have finite transition bands.



Hilbert transform : Time - domain transformation

- transform a signal from time domain to another time domain to get another signal.



$$h_{HT}(t) \xleftrightarrow{\text{F.T.}} H_{HT}(f)$$

$\therefore$ $$\boxed{\hat{m}(t) = m(t) \circledast h_{HT}(t)}$$

convolution

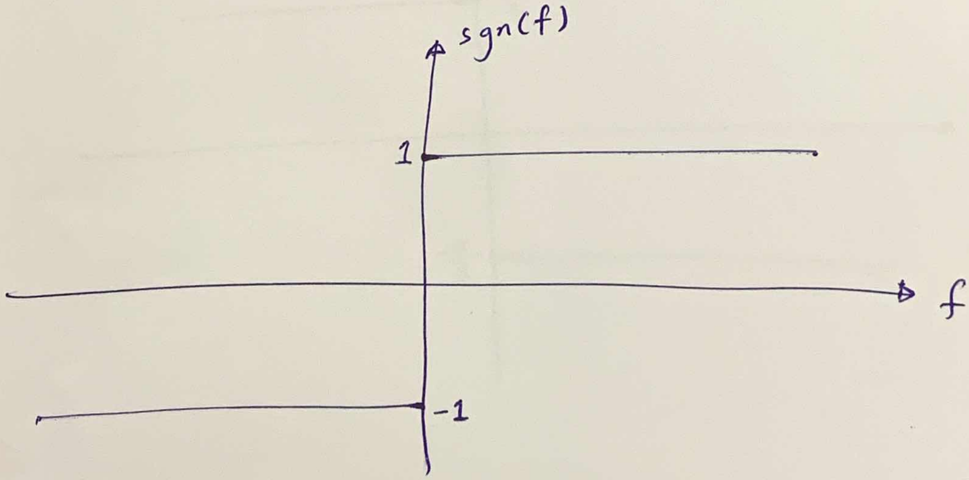
Find expression for $h_{HT}(t)$ to characterize H.T.

- However, it is easier to characterize the spectrum of $H_{HT}(f)$.

$$\boxed{H_{HT}(f) = -J\,\mathrm{sgn}(f)}$$

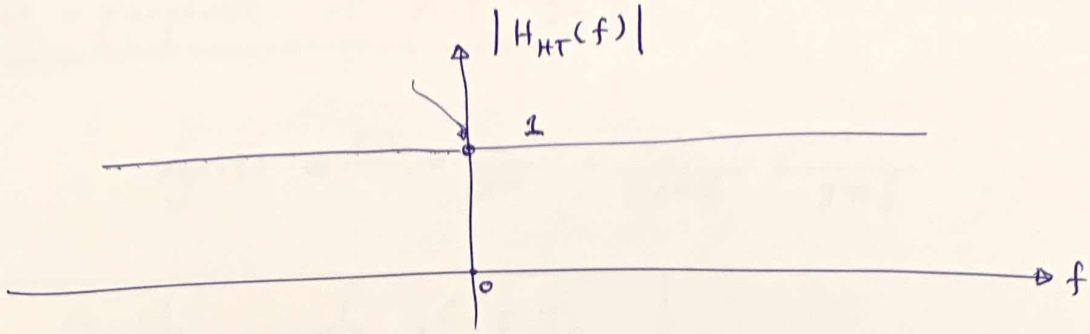
where $\mathrm{sgn}(f) = \begin{cases} 1 & f>0 \\ 0 & f=0 \\ -1 & f<0 \end{cases}$

$$\Rightarrow \boxed{H_{HT}(f) = \begin{cases} -J & f>0 \\ 0 & f=0 \\ J & f<0 \end{cases}}$$

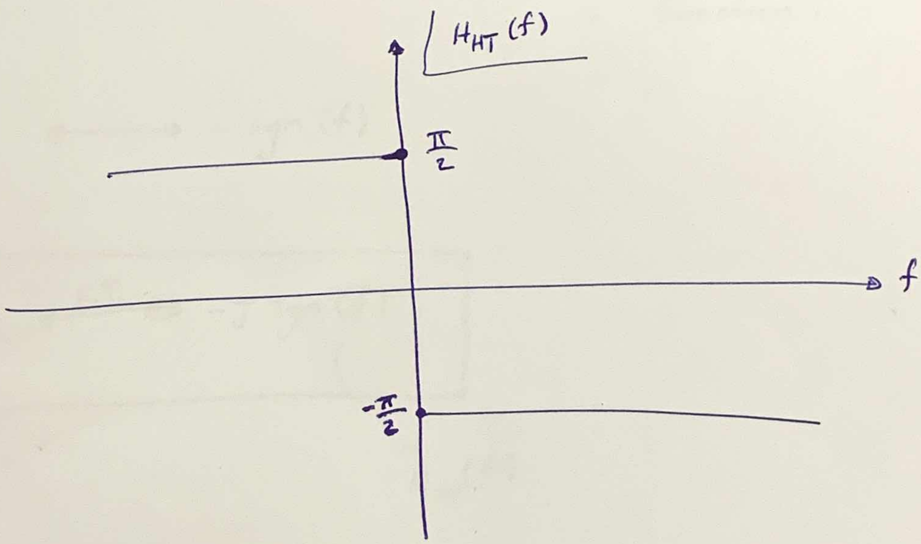


$$|H_{HT}(f)| = |-J\,\mathrm{sgn}(f)| = |-J|\cdot|\mathrm{sgn}(f)| = 1\cdot\underbrace{|\mathrm{sgn}(f)|}_{\substack{1,\ f\neq 0 \\ 0,\ f=0}}$$

$$= \begin{cases} 1 & ,\ f\neq 0 \\ 0 & ,\ f=0 \end{cases}$$



$$\angle H_{HT}(f) = \angle -j\,\mathrm{sgn}(f) = \begin{cases} -\frac{\pi}{2} & f > 0 \\ 0 & f = 0 \\ \frac{\pi}{2} & f < 0 \end{cases}$$



- We observe it does shifting the phase b/c gain is unity.

So it is phase shifter.

All +ve frequencies by $-\frac{\pi}{2}$.

All -ve frequencies by $\frac{\pi}{2}$.

Impulse - response of H.T. :-

$$\text{sgn}(t) \xleftrightarrow{\text{F.T.}} \frac{2}{J\omega} = \frac{\cancel{2}}{J\cancel{2}\pi f} = \frac{1}{J\pi f}$$

$$\boxed{\begin{array}{l}\text{using Duality - property of F.T.} \\ \qquad x(t) \longleftrightarrow X(f) \\ \qquad X(t) \longleftrightarrow x(-f)\end{array}}$$

$$\therefore \quad \frac{1}{J\pi t} \longleftrightarrow \text{sgn}(-f)$$

; $\because$ sgn function is odd function.

$$\frac{1}{J\pi t} \longleftrightarrow -\text{sgn}(f)$$

$$\therefore \quad \boxed{\underbrace{\frac{1}{\pi t}}_{h_{HT}(t)} \xleftrightarrow{\text{F.T.}} \underbrace{-J\,\text{sgn}(f)}_{H_{HT}(f)}}$$

$$\therefore \quad \boxed{h_{HT}(t) = \frac{1}{\pi t}}$$

$$m(t) \longrightarrow \boxed{H.T.} \longrightarrow \hat{m}(t)$$

$$\hat{m}(t) = m(t) \circledast h_{HT}(t)$$

$$\boxed{\therefore \quad \hat{m}(t) = m(t) \circledast \frac{1}{\pi t}}$$ time-domain representation.

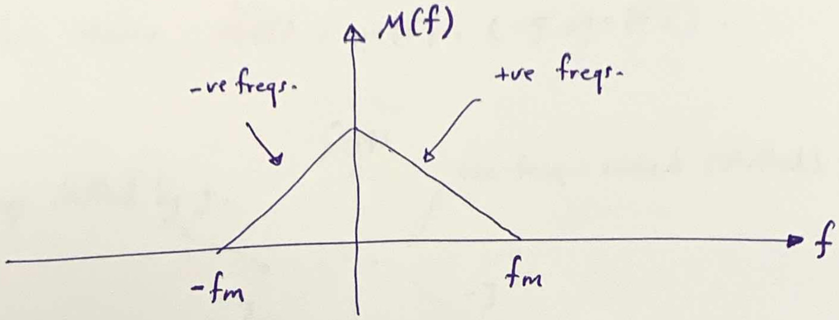
$$\hat{M}(f) = M(f) \cdot H_{HT}(f)$$

$$= M(f) \cdot -J\,\mathrm{sgn}(f)$$

$$\boxed{\therefore \quad \hat{M}(f) = -J\,\mathrm{sgn}(f) \cdot M(f)}$$ frequency-domain representation.

SSB Generation using phase-shift Method :-

- consider message signal $m(t)$ with spectrum $M(f)$.



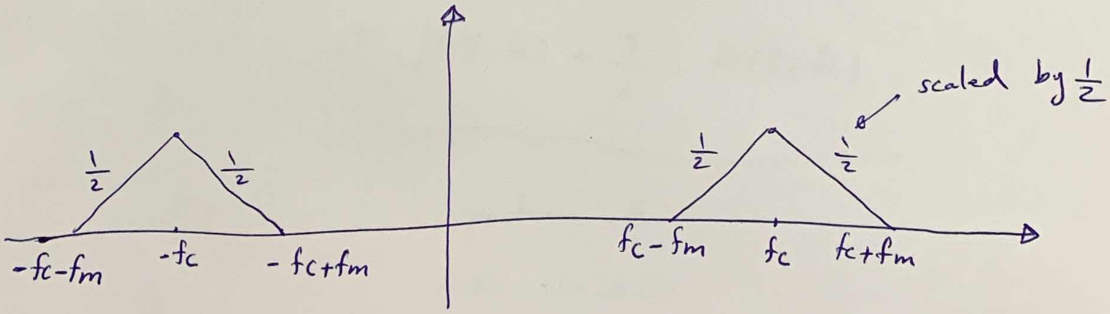
- consider the modulated signal $x(t)$ given as :-

$$x(t) = m(t) \cdot \cos 2\pi f_c t - \hat{m}(t) \sin 2\pi f_c t$$, where

$\hat{m}(t)$ is the Hilbert transform of $m(t)$.

- then we will show that $x(t)$ is a SSB signal.

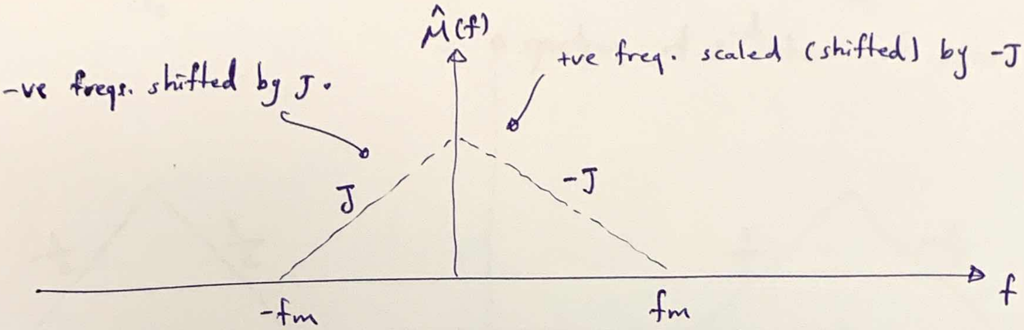
$$m(t) \cdot \cos 2\pi f_c t \longleftrightarrow \frac{1}{2} M(f-f_c) + \frac{1}{2} M(f+f_c)$$



$\hat{m}(t) \cdot \sin 2\pi f_c t \longleftrightarrow$

$\hat{m}(t) \longleftrightarrow \hat{M}(f)$

we Know $\hat{M}(f) = M(f) \cdot (-j \, \mathrm{sgn}(f))$.



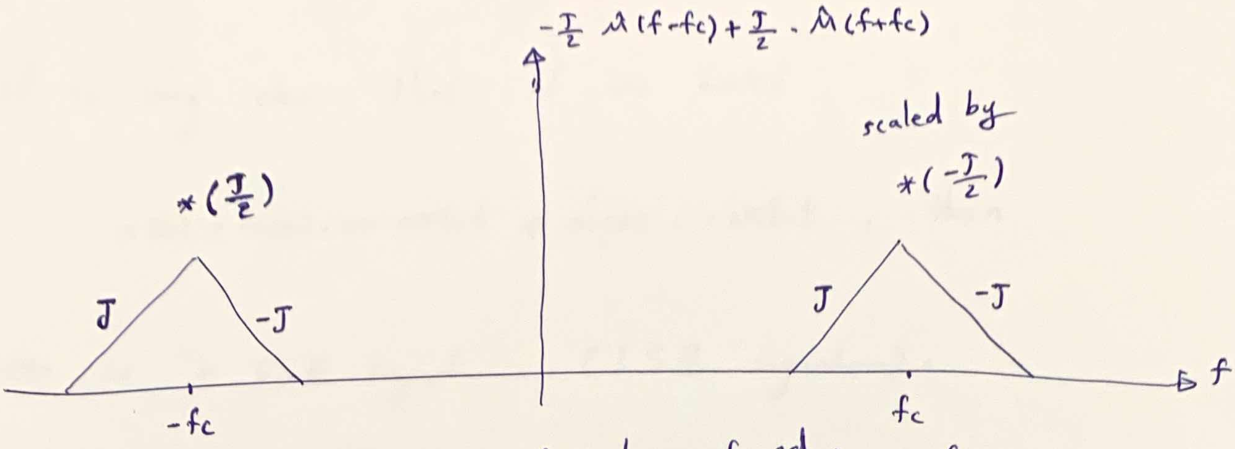
$$\hat{m}(t) \cdot \sin 2\pi f_c t = \hat{m}(t) \cdot \left(\frac{e^{j2\pi f_c t} - e^{-j2\pi f_c t}}{2j} \right)$$

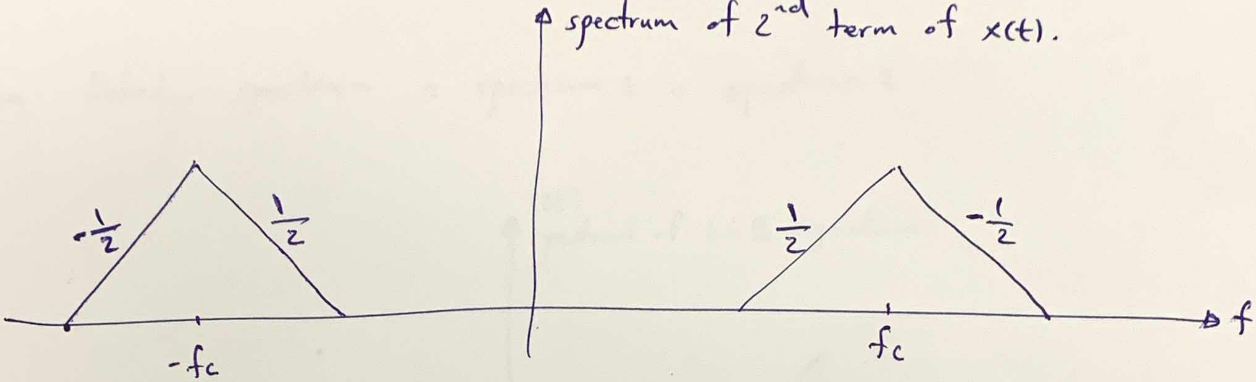
$$= \frac{\hat{m}(t) \cdot e^{j2\pi f_c t}}{2j} - \frac{\hat{m}(t) \cdot e^{-j2\pi f_c t}}{2j}$$

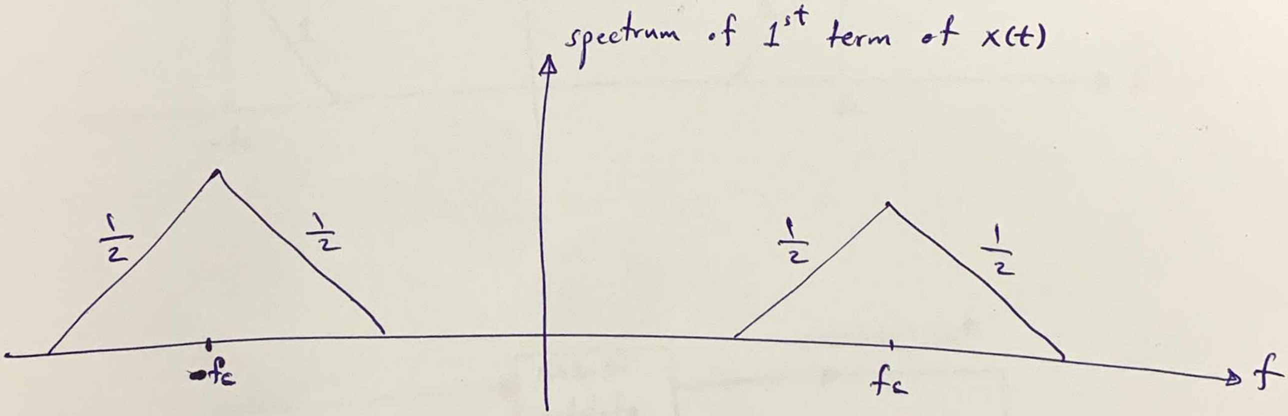
$\Updownarrow$ modulation property. F.T. $\qquad \Updownarrow$ F.T.

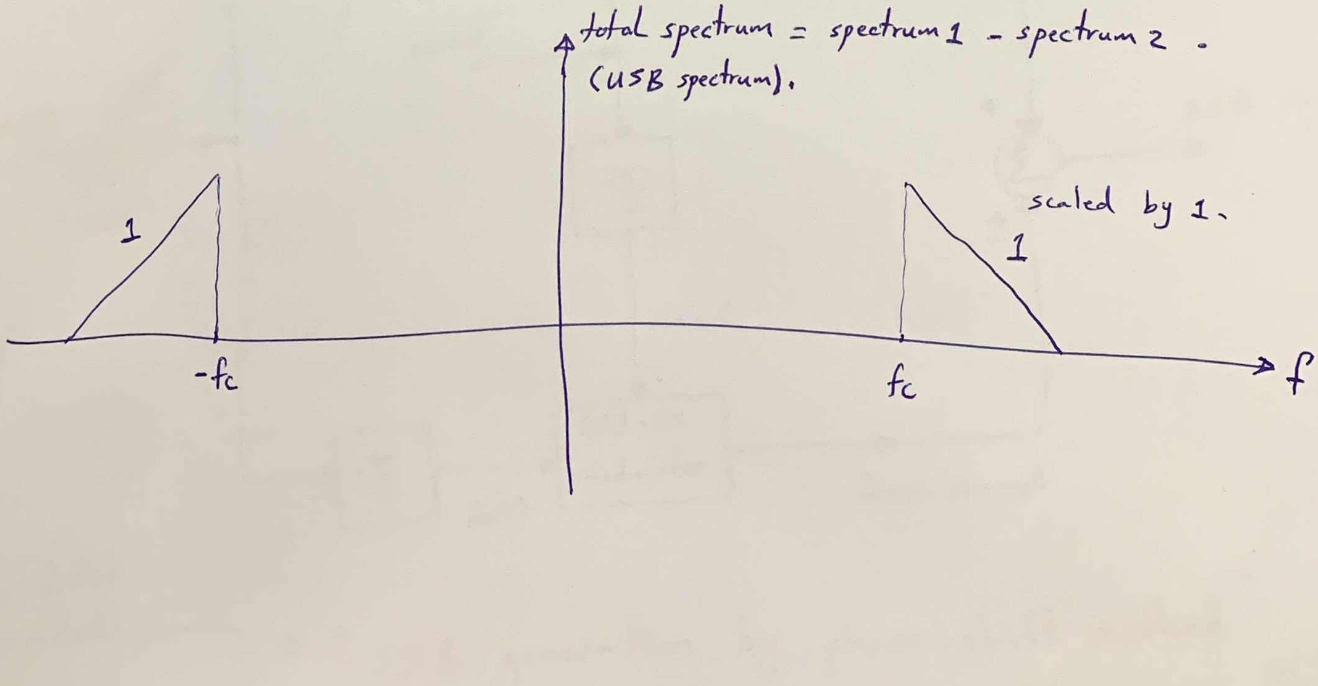
$$= \frac{\hat{M}(f-f_c)}{2j} - \frac{\hat{M}(f+f_c)}{2j}$$

$$= \underbrace{\frac{-j}{2} \cdot \hat{M}(f-f_c) + \frac{j}{2} \cdot \hat{M}(f+f_c)}_{\text{spectrum of } \hat{m}(t) \cdot \sin 2\pi f_c t}$$


$-\frac{J}{2}\,\hat{m}(f-f_c) + \frac{J}{2}\cdot\hat{m}(f+f_c)$
scaled by
$*(-\frac{J}{2})$
$*(\frac{J}{2})$
J
-J
J
-J
$-f_c$
f_c
f


spectrum of 2^{nd} term of x(t).
$-\frac{1}{2}$
$\frac{1}{2}$
$\frac{1}{2}$
$-\frac{1}{2}$
$-f_c$
f_c
f


spectrum of 1^{st} term of x(t)
$\frac{1}{2}$
$\frac{1}{2}$
$\frac{1}{2}$
$\frac{1}{2}$
$-f_c$
f_c
f

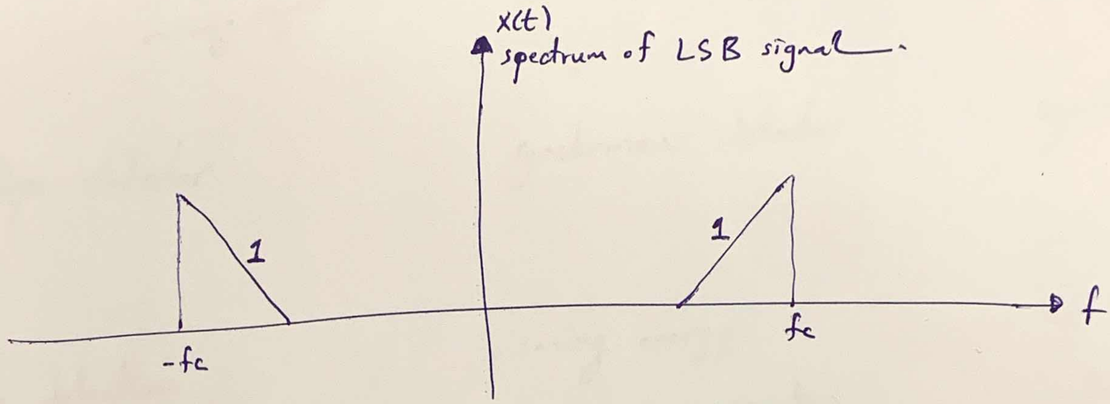

total spectrum = spectrum 1 - spectrum 2 .
(USB spectrum).
scaled by 1.
1
1
$-f_c$
f_c
f

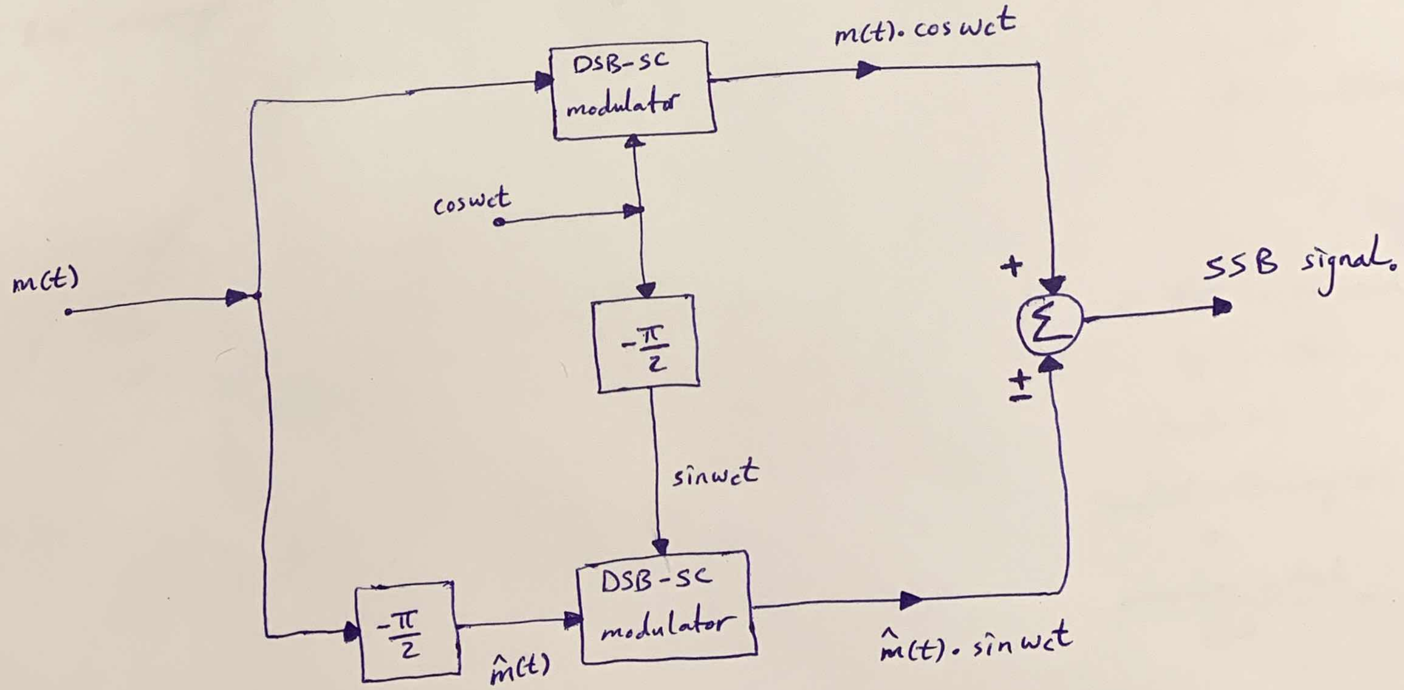
- it is very clear that if we make

$$x(t) = m(t) \cdot \cos 2\pi f_c t + \hat{m}(t) \sin 2\pi f_c t \quad , \text{ then}$$

$x(t)$ is a SSB signal (LSB signal).

then total spectrum = spectrum 1 + spectrum 2.





- no filter is needed. → " SSB generation by phase-shift method "

AM wave

$P = P_c + P_s$

$BW = 2B$

max. freq. of message signal

envelope detector

- simple detection ckt. with high power & BW consumption

DSB-SC

$P = P_s$

$BW = 2B$

synchronous detector

- saving energy
- high B.W consumption
- synchronization problem

SSB-SC

$P = \frac{P_s}{2}$

$BW = B$

synchronous detector

- saving energy & BW.
- synchronization & sharping cut-off filter problem

$\because$ BW is reduced by $\frac{1}{2}$. This leads to

spectral efficiency (SE) $= \frac{\text{information content}}{\text{B.W}}$

$\therefore$ SE is doubled.

b/c information in SSB is same for DSB.